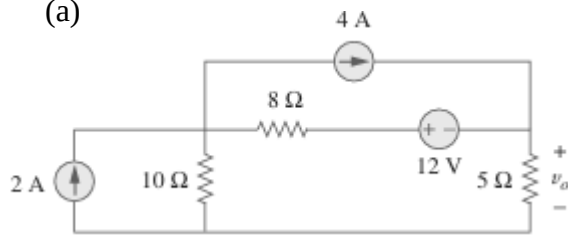
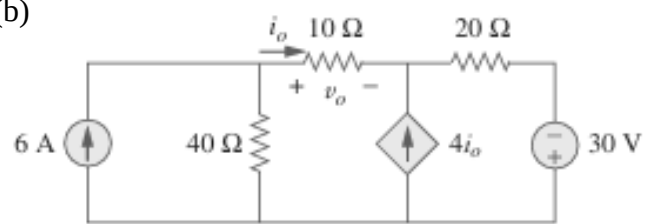


1. Find v_o using superposition theorem.

(a)

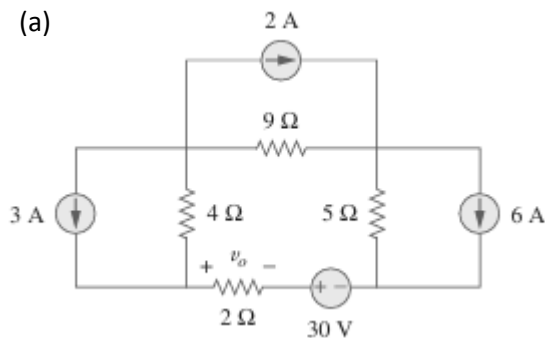


(b)

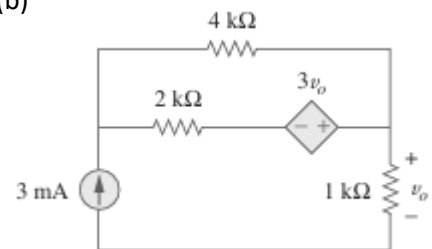


2. Find v_o using source transformation

(a)

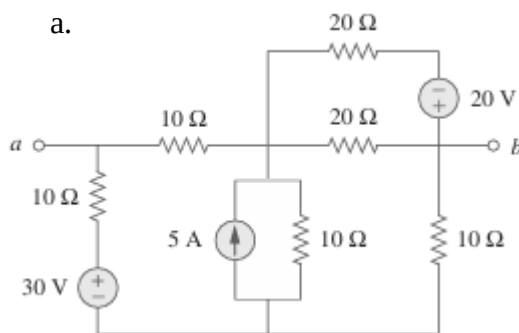


(b)

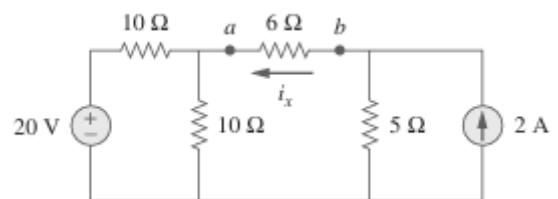


3. Find the Thevenin's equivalent circuit at the terminals a and b .

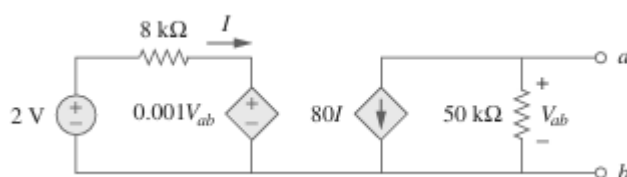
a.



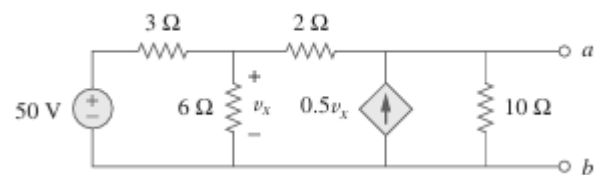
b.



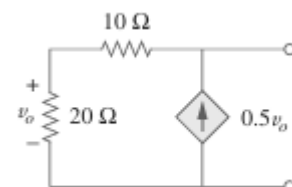
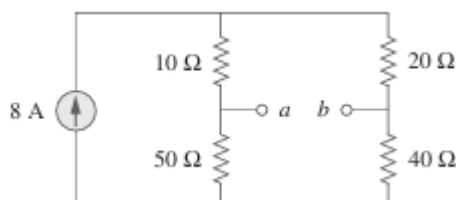
c.



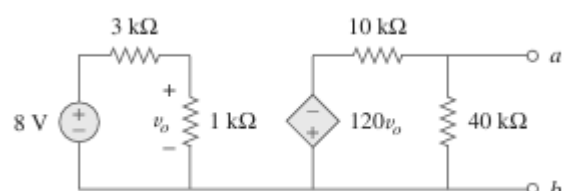
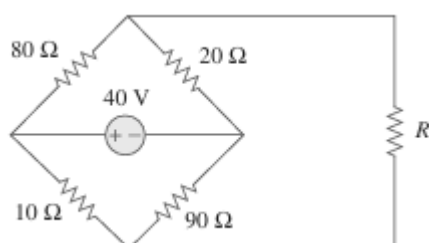
d.



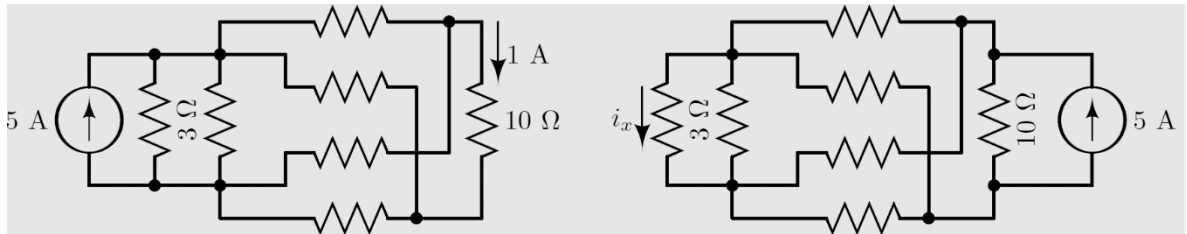
4. Find the Norton's equivalent circuit



5. Find the value R for maximum power transfer. Also determine the maximum power. In the second circuit, R is connected between the terminals a and b .



6. In the circuit shown below, the values of the unmarked are unknown. When a current of 5 A is applied in the schematic on the left, a current of 1 A is measured through the $10\ \Omega$ resistor. What will be i_x , the current in the $3\ \Omega$ resistor when a current of 5 A is applied as in the schematic on the right?



7. Verify Tellegen's for the circuit shown in figure below.

